

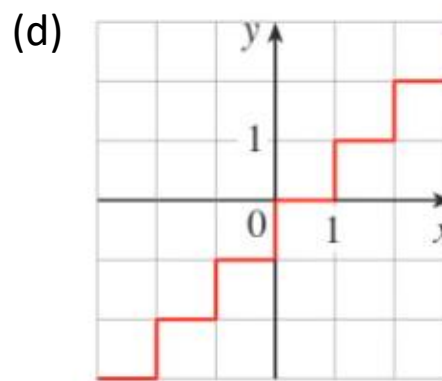
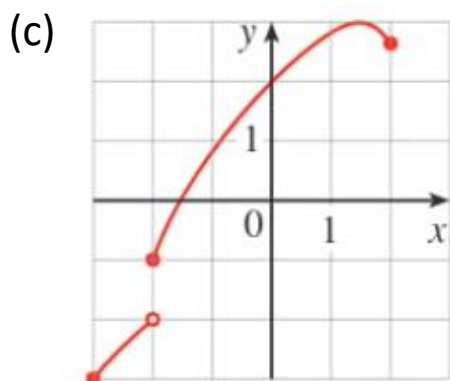
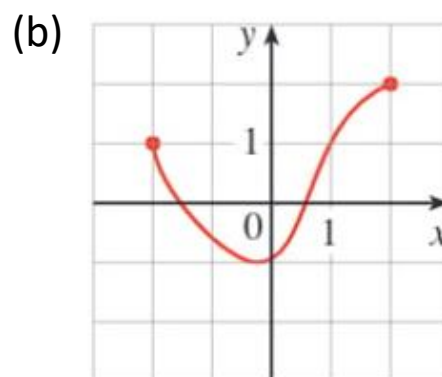
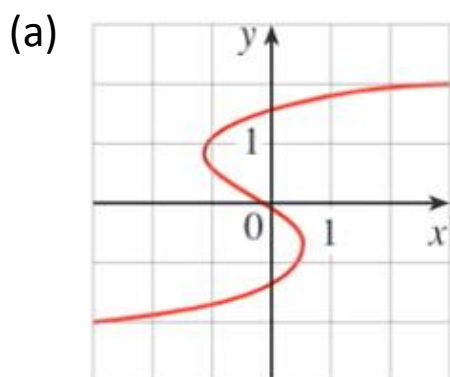
CSD1251 Week 2 Tutorial Problems

12th – 18th January 2026

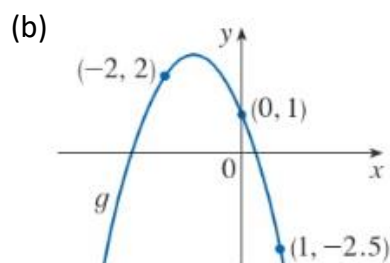
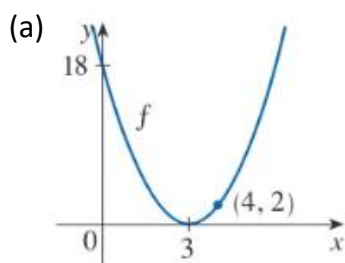
It is recommended to treat the attempt of these problems seriously, even though they are not graded. You may refer to the lecture slides if you are unsure of any concepts.

After attempting each problem, think about what you have learned from the attempt as a means of consolidating what you have learned.

Question 1. Determine if the curve in each of the following parts is the graph of a function of x . If it is, state the domain and range of the function.



Question 2. The following graphs below are graphs of quadratic functions f and g , along with some of their points. Find f and g .



Question 3. Complete the square for each of the following functions.

- (a) $f(x) = x^2 + 4x + 6$ (b) $g(x) = x^2 - 8x - 49$ (c) $h(x) = x^2 - 12x + 169$
 (d) (e) $q(x) = x^4 - 2x^2 + 3$ $u(x) = y^4 - 6y^2 + 3$

Question 4. For each of the following pairs of functions, find $f \circ g$ and $g \circ f$.

- (a) $f(x) = \tan x$, $g(x) = 1 + x^2$ (b) $f(x) = \sin x$, $g(x) = 1 - x^2$
 (c) $f(x) = \ln x$, $g(x) = e^{2x}$ (d) $f(x) = \sec x$, $g(x) = x^2 - 1$.

Question 5. For the following functions and their given domains (if they are not given, take the largest possible domain), check for each of them, if the inverse exists. If they do exist, then find it.

- (a) $f(x) = x^4 + 3$, $x \in \mathbb{R}$
 (b) $g(x) = x^4 + 3$, $x \in (-\infty, 0]$
 (c) $h(x) = 1 - \sin x$

Question 6. The inverse for each of these functions is guaranteed to exist. Find the inverse of each of these functions.

- (a) $q(x) = (x + 6)^2 - 2$, $x \leq -6$
 (b) $f(x) = x^2 + 4x$, $x \geq -2$
 (c) $g(x) = \frac{6 - 3x}{5x + 7}$
 (d) $h(x) = \frac{1 - e^{-x}}{1 + e^{-x}}$

Final Answers:

Q1(a) Not a function; (b) Function, domain = $[-2, 2]$, range = $[-1, 2]$; (c) Function, domain = $[-3, 2]$, range = $[-3, -2] \cup [-1, 3]$ or $[-3, 3] \setminus [-2, -1]$; (d) Not a function.

Q2(a) $f(x) = 2x^2 - 12x + 18$; (b) $g(x) = -x^2 - 2.5x + 1$.

Q3(a) $(x + 2)^2 + 2$, (b) $(x - 4)^2 - 65$, (c) $(x - 6)^2 + 133$, (d) $(x^2 - 1)^2 + 2$, (e) $(y^2 - 3)^2 - 6$.

Q4(a) $(f \circ g)(x) = \tan(1 + x^2)$, $(g \circ f)(x) = \sec^2 x$, (b) $(f \circ g)(x) = \sin(1 - x^2)$, $(g \circ f)(x) = \cos^2 x$, (c) $(f \circ g)(x) = 2x$, $(g \circ f)(x) = x^2$, $(f \circ g)(x) = \sec(x^2 - 1)$, $(g \circ f)(x) = \tan^2 x$.

Q5(a) No inverse, (b) $g^{-1}(x) = -\sqrt[4]{x - 3}$, (c) No inverse.

Q6(a) $q^{-1}(x) = -6 - \sqrt{x + 2}$, (b) $f^{-1}(x) = \sqrt{x + 4} - 2$, (c) $g^{-1}(x) = \frac{6 - 7x}{5x + 3}$,

(d) $h^{-1}(x) = \ln\left(\frac{1 + x}{1 - x}\right)$.